

Introduction

Bird sound classification is an important application of deep learning that can support wildlife monitoring, biodiversity research, and environmental analysis. Since bird calls contain unique sound patterns, audio recordings can be converted into spectrograms, which visually represent sound frequencies over time. These spectrograms can then be analyzed using Convolutional Neural Networks (CNNs), a deep learning model commonly used for image classification tasks because of its ability to automatically learn complex spatial patterns from data.

The goal of this research is to investigate how accurately CNNs can classify bird species using spectrogram images generated from bird audio recordings. Two classification tasks are explored throughout this research. The first task focuses on multi-class classification using 12 Seattle-area bird species. The second task investigates binary classification using only two bird species, *amerob* and *houspa*, in order to compare performance between a simpler binary problem and a more difficult multi-class problem. In addition, the trained multi-class model is used to predict bird species from three external mp3 audio clips that may contain overlapping bird calls.

The dataset used in this research consists of bird audio recordings that were converted into mel spectrogram images before training the models. Each spectrogram represents the frequency intensity of bird calls over time and has dimensions of 128 frequency bins by 517 time steps. The full dataset contains 1,981 spectrogram samples distributed across 12 bird species, although the number of samples varies significantly between classes, creating class imbalance within the dataset. Along with the spectrogram data stored in an HDF5 file, metadata files containing species labels and recording information are also used during preprocessing and model training. Multiple CNN architectures are explored throughout this research, including a baseline CNN model and improved CNN models designed to reduce overfitting and improve classification performance.

Theoretical Background

This research uses Convolutional Neural Networks (CNNs), a deep learning method designed for image-based pattern recognition tasks. CNNs are especially useful for spectrogram classification because spectrograms visually represent sound frequencies over time, allowing the network to learn spatial and frequency-based patterns directly from the data. Instead of manually extracting sound features, CNNs automatically learn important patterns during training through convolutional filters and feature maps.

The primary layers used in the CNN architectures include convolutional layers, pooling layers, dense layers, and dropout layers. Convolutional layers apply small filters across the spectrogram images to detect important local patterns such as edges, harmonics, and repeating sound structures within bird calls. ReLU activation functions are used after each convolutional layer because they introduce nonlinearity while remaining computationally efficient. Max pooling layers are then used to reduce the spatial dimensions of the feature maps, which lowers computational cost and helps reduce overfitting by keeping only the most important features.

After feature extraction, the feature maps are flattened into one-dimensional vectors before being passed into dense layers for classification. Dense layers combine the extracted features and perform the final prediction task. Dropout regularization is also included in several models to reduce overfitting by randomly disabling neurons during training. For the multi-class classification task involving 12 bird species, the output layer uses a softmax activation function to produce probability distributions across all species classes. For the binary classification task, a sigmoid activation function is used because the model only predicts between two possible classes.

Several CNN architectures are explored throughout this research. The baseline CNN model uses two convolutional layers followed by max pooling, flattening, and dense classification layers. Additional improved architectures are also tested by increasing the number of convolutional layers and adding regularization techniques such as dropout, batch normalization, and early stopping. Batch normalization is intended to stabilize training and improve convergence by normalizing intermediate activations, while early stopping helps reduce overfitting by stopping training once validation loss no longer improves.

Multiple hyperparameters are adjusted throughout model training, including the number of epochs, batch size, number of convolutional filters, dropout rate, and network depth. Epochs determine how many times the model sees the full training dataset, while batch size controls how many samples are processed before updating model weights. The Adam optimizer is used for gradient-based optimization because it generally performs well on deep learning classification tasks. For evaluation, the models are assessed using accuracy, loss, confusion matrices, precision, recall, and F1-scores. Accuracy measures the percentage of correctly classified samples, while loss measures prediction error during optimization. Confusion matrices are particularly useful for identifying which bird species are most commonly confused by the model.

Overfitting is also an important concept explored throughout this research. Overfitting occurs when a model memorizes the training data instead of learning generalized patterns that transfer well to unseen data. This can often be identified when training accuracy continues increasing while validation accuracy stagnates or validation loss increases. Several experiments performed during this research demonstrate how increasing model complexity does not always improve performance, particularly when working with imbalanced datasets and limited training samples.

Methodology

Data Preparation

The dataset used in this research consisted of bird audio recordings that were converted into mel spectrogram images and stored in an HDF5 file. The dataset contained 1,981 spectrogram samples representing 12 Seattle-area bird species. Each spectrogram had dimensions of 128 frequency bins by 517 time steps. Species labels were obtained from the accompanying metadata files and used as the target variable for classification.

The spectrogram data for each species was loaded from the HDF5 file and combined into a single dataset. The resulting feature matrix contained 1,981 spectrogram images, while the label vector contained the corresponding bird species labels. The labels were converted from text values into numerical class labels using LabelEncoder. Spectrogram values were normalized to a range between 0 and 1 to improve training stability and model performance. The data was then reshaped to include a channel dimension, resulting in an input shape of (128, 517, 1) for each sample.

The multi-class dataset was divided into training and testing sets using an 80/20 split. Stratified sampling was used to preserve the original class distribution across both sets. This resulted in 1,584 training samples and 397 testing samples.

Multi-Class CNN Models

Several CNN architectures were explored throughout this research. The baseline CNN consisted of two convolutional layers with ReLU activation functions and max pooling layers, followed by a flatten layer, a dense layer with 64 neurons, a dropout layer with a rate of 0.3, and a 12-node softmax output layer for multi-class classification. The model was trained using the Adam optimizer and sparse categorical cross-entropy loss function for 10 epochs with a batch size of 32.

Additional architectures were tested to investigate whether increased model complexity would improve performance. One experimental model included three convolutional layers, batch normalization layers, and additional dense layers. However, this model experienced unstable training behavior and rapidly increasing validation loss. A second improved architecture retained the additional convolutional layer while removing batch normalization and increasing dropout regularization. Early stopping was implemented using validation loss monitoring with a patience value of three epochs to reduce overfitting and restore the best-performing model weights.

Model performance was evaluated using test accuracy, test loss, confusion matrices, classification reports, and training history plots. Accuracy and loss curves were used to monitor learning progress and identify signs of overfitting.

Binary CNN Model

A separate binary classification experiment was conducted using the species *amerob* and *houspa*. Because *houspa* contained substantially more samples than *amerob*, a random subset of 172 *houspa* samples was selected to create a balanced dataset. This resulted in 344 total spectrogram samples, with 172 observations from each species.

The binary dataset was normalized using the same preprocessing procedure as the multi-class dataset and then divided into training and testing sets using an 80/20 stratified split. The resulting training set contained 275 samples and the testing set contained 69 samples.

The binary CNN architecture consisted of two convolutional layers with max pooling, followed by a flatten layer, a dense layer with 64 neurons, a dropout layer with a rate of 0.3, and a single sigmoid output node. The model was trained using binary cross-entropy loss and the Adam optimizer for 10 epochs with a batch size of 16. Model performance was evaluated using test accuracy, test loss, confusion matrices, classification reports, and training history plots.

External Audio Classification

To evaluate model performance on unseen data, three external mp3 bird recordings were provided. Each audio clip was converted into a mel spectrogram using the same preprocessing methodology used to create the training dataset. The spectrograms were normalized and passed through the trained multi-class CNN model to generate species predictions and prediction confidence scores. Spectrogram visualizations were also examined to identify clips that potentially contained multiple bird calls based on overlapping frequency patterns and model uncertainty.

Results

4.1 Multi-Class CNN Results

Figure 1 shows the final performance comparison for the two multi-class CNN models. The Baseline CNN achieved a test accuracy of 33.00% with a test loss of 2.0173. An initial improved architecture that included batch normalization experienced unstable training behavior, causing the validation loss to increase rapidly while validation accuracy remained low. Because the model failed to converge properly, it was excluded from further analysis. The Stable Improved CNN performed better, reaching 42.57% accuracy with a lower test loss of 1.8165. It also improved across macro and weighted precision, recall, and F1-score, showing that it performed better across the 12 bird species overall.

Figure 1. Multi-Class CNN Performance Comparison

	Model	Accuracy	Loss	Macro Precision	Macro Recall	Macro F1	Weighted Precision	Weighted Recall	Weighted F1
0	Baseline CNN	0.3300	2.0173	0.1055	0.0961	0.0658	0.2106	0.3300	0.1920
1	Stable Improved CNN	0.4257	1.8165	0.3201	0.2586	0.2532	0.3980	0.4257	0.3826

4.2 Stable CNN Training Performance

Training accuracy increased steadily throughout the learning process, rising from approximately 30% during the first epoch to over 73% by the final epoch. Validation accuracy also improved during training, increasing from roughly 32% to a peak of about 44% before leveling off. The gap between training and validation accuracy became larger during the later epochs, suggesting that the model continued to learn patterns from the training data while achieving more modest improvements on unseen data.

The loss curves provide additional insight into model performance. Training loss decreased consistently from approximately 2.28 to 0.86, indicating that the model was successfully fitting the training data. Validation loss decreased during the early epochs and reached its lowest value of approximately 1.82 around epoch 6. After this point, validation loss began to increase while training loss continued to decrease, indicating the onset of overfitting. Early stopping helped limit this effect by restoring the model weights corresponding to the best validation performance.

Overall, the training history shows that the Stable Improved CNN learned meaningful spectrogram features and achieved better generalization performance than the Baseline CNN. Although some overfitting was observed in the later epochs, the model achieved its strongest performance near epoch 6, where validation loss was minimized and validation accuracy remained relatively high.

Figure 2: Stable CNN Accuracy Over Epochs

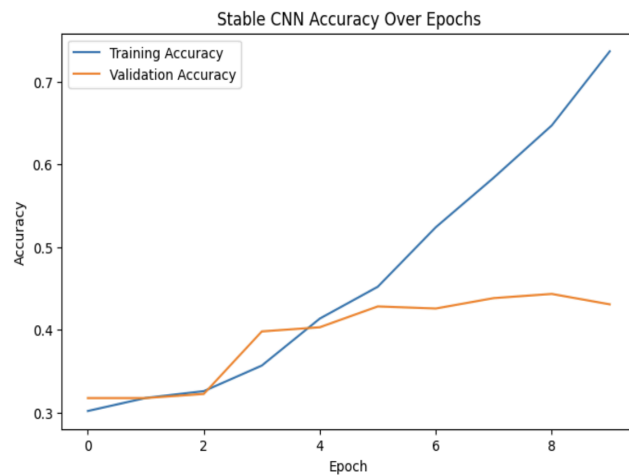
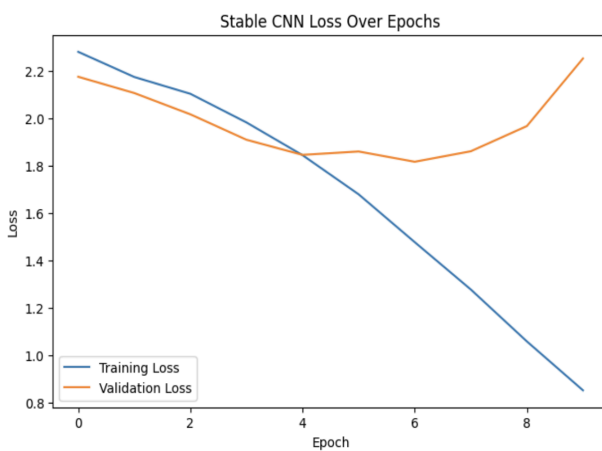


Figure 3: Stable CNN Loss Over Epochs



4.3 Multi-Class Classification Results

Figure 4: Stable CNN Confusion Matrix

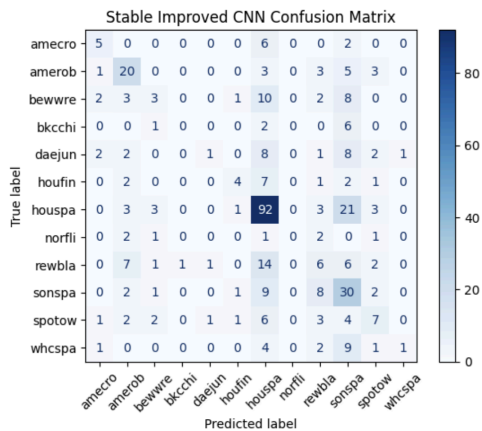


Figure 5: Stable CNN Classification Report

	precision	recall	f1-score	support
amecro	0.42	0.38	0.40	13
amerob	0.47	0.57	0.51	35
bewwre	0.25	0.10	0.15	29
bkcchi	0.00	0.00	0.00	9
daejun	0.33	0.04	0.07	25
houfin	0.50	0.24	0.32	17
houspa	0.57	0.73	0.64	126
norfli	0.00	0.00	0.00	7
rewbla	0.19	0.16	0.17	38
sonspa	0.30	0.57	0.39	53
spotow	0.32	0.26	0.29	27
whcspa	0.50	0.06	0.10	18
accuracy			0.43	397
macro avg	0.32	0.26	0.25	397
weighted avg	0.40	0.43	0.38	397

The confusion matrix and classification report provide a more detailed view of the Stable Improved CNN’s performance across the 12 bird species. Overall test accuracy reached 42.57%, which represents an improvement over the Baseline CNN. Performance varied substantially across classes. The model performed best on houspa, achieving a precision of 0.57, recall of 0.73, and F1-score of 0.64. Moderate performance was also observed for amerob and sonspa. However, several species such as bkcchi and norfli achieved near-zero precision and recall, indicating that the model struggled to learn distinguishing features for these classes.

The confusion matrix shows that many classification errors occurred between species with similar spectrogram patterns. In particular, the model frequently predicted houspa when classifying other bird species, suggesting that the acoustic characteristics of houspa were easier for the network to recognize than those of several less represented classes. Despite these limitations, the Stable Improved CNN produced more balanced predictions than the Baseline CNN and demonstrated improved generalization across multiple evaluation metrics.

4.4 Binary CNN Results

To evaluate performance on a simpler classification task, a binary CNN was trained using recordings from *amerob* and *houspa*. The dataset was balanced to include 172 recordings from each species, resulting in 344 total samples. After preprocessing and train-test splitting, the model was trained for ten epochs.

Figure 6: Binary CNN Accuracy Over Epochs

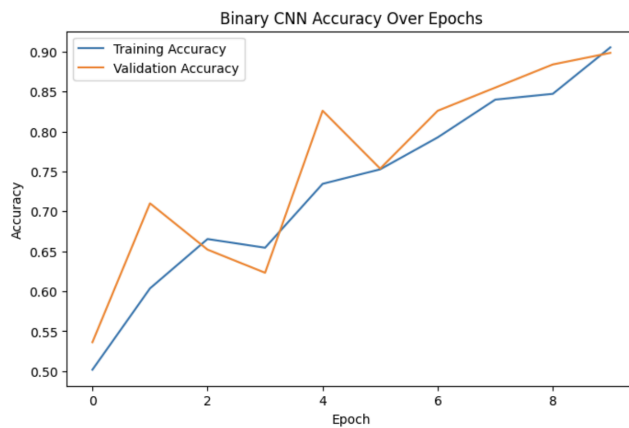


Figure 6 shows the training and validation accuracy throughout the training process. Training accuracy increased steadily from approximately 50.3% during the first epoch to 90.6% by the final epoch. Validation accuracy also improved throughout training, reaching approximately 90.0% by the final epoch. The relatively small gap between the training and validation curves suggests that the model generalized well to unseen data.

Insert Figure 7: Binary CNN Loss Over Epochs

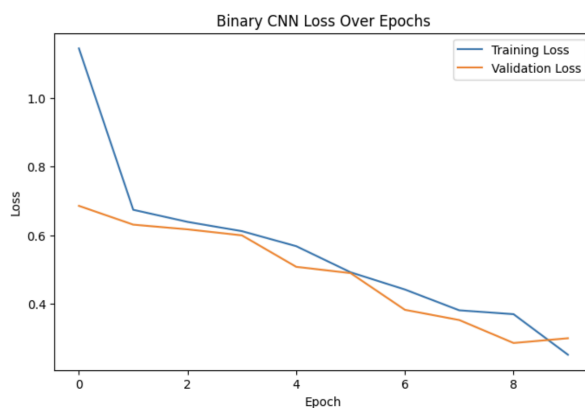


Figure 7 presents the training and validation loss curves. Training loss decreased from approximately 1.14 during the first epoch to 0.25 by the final epoch, while validation loss decreased from approximately 0.69 to 0.30. Both curves followed a similar downward trend, indicating stable learning and effective convergence throughout training.

The final binary CNN achieved a test accuracy of 89.86% with a test loss of 0.2998. These results demonstrate that the binary classification task was substantially easier than the multi-class classification problem, allowing the model to learn more discriminative features for distinguishing between the two bird species.

4.5 Binary CNN Confusion Matrix

Figure 8: Binary CNN Confusion Matrix and Classification Report

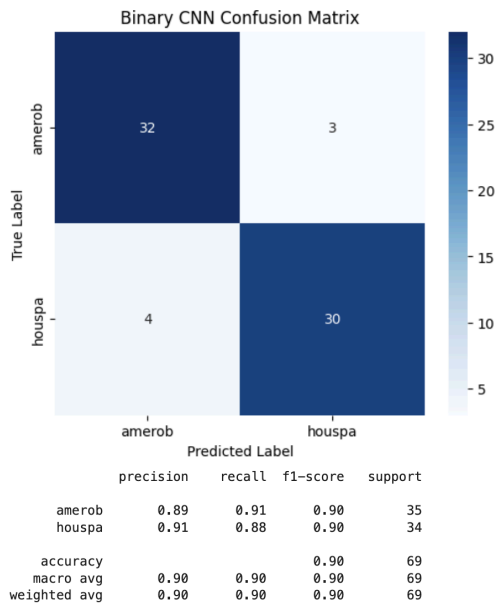


Figure 8 presents the confusion matrix and classification report for the binary CNN model. The model correctly classified 32 of 35 amerob recordings and 30 of 34 houspa recordings. Three amerob recordings were incorrectly classified as houspa, while four houspa recordings were incorrectly classified as amerob. Overall, the confusion matrix shows that the model performed well on both classes and produced relatively balanced predictions.

The classification report further demonstrates the model’s strong performance. For amerob, the model achieved a precision of 0.89, recall of 0.91, and F1-score of 0.90. For houspa, the model achieved a precision of 0.91, recall of 0.88, and F1-score of 0.90. The overall accuracy was approximately 90%, with both the macro-average and weighted-average F1-scores equal to 0.90. These results indicate that the model was able to distinguish between the two species with a high degree of consistency.

4.6 External Audio Predictions

The final stable CNN model was used to classify three external audio recordings that were not part of the original training dataset. Each audio file was converted into a spectrogram using the same preprocessing pipeline that was used during model training

Figure 9: Spectrograms of External Audio Recordings

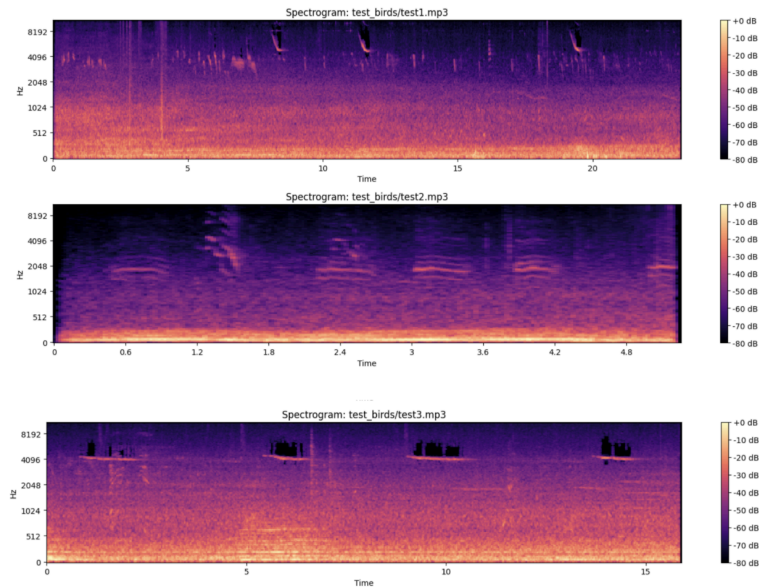


Figure 9 shows the spectrograms generated from the three external audio clips. Visual inspection suggests that the recordings contain different acoustic patterns and varying levels of background noise. Some recordings contain multiple sound structures occurring at different frequencies and time intervals, which may make classification more challenging than the training examples.

Figure 10: Table of External Audio Predictions

	Audio Clip	Predicted Species	Confidence
0	test1.mp3	houspa	0.3752
1	test2.mp3	houspa	0.6421
2	test3.mp3	houspa	0.2834

As shown in figure 10, the model predicted all three recordings as houspa. The confidence scores ranged from 0.2834 to 0.6421, indicating varying levels of certainty across the predictions. The highest confidence prediction was test2.mp3 with a confidence score of 0.6421, while test3.mp3 produced the lowest confidence score of 0.2834.

The relatively low confidence values suggest that the model was uncertain when classifying these previously unseen recordings. This uncertainty may be caused by background noise, overlapping bird calls, or differences between the external recordings and the training dataset. Although all three clips were classified as houspa, the confidence scores indicate that these predictions should be interpreted cautiously.

Discussion

This research explored the use of convolutional neural networks (CNNs) to classify bird species from spectrogram images generated from bird call recordings. Two classification tasks were investigated: a multi-class classification problem involving all 12 bird species and a binary classification problem involving amerob and houspa.

The multi-class classification task proved substantially more difficult than the binary classification task. The baseline CNN achieved a test accuracy of 33.0%, while the final stable CNN improved performance to 42.57%. Although this represents a meaningful improvement, the model still struggled to distinguish among all twelve species. This outcome is not surprising because many bird species produce calls with similar acoustic characteristics, making it difficult for the model to learn highly discriminative features from a relatively limited dataset.

One challenge encountered during this research was model instability. An attempted improved CNN architecture that incorporated batch normalization produced unstable validation performance and failed to generalize effectively. As a result, the architecture was modified and a simpler stable CNN was developed, which produced more reliable performance. Training time remained manageable throughout the project. The stable multi-class CNN trained for approximately ten epochs before early stopping restored the best-performing model weights. The binary CNN trained even more efficiently while achieving substantially stronger results.

The binary classification task achieved significantly higher performance than the multi-class task. The binary CNN reached a test accuracy of 89.86%, demonstrating that the model was much more successful when distinguishing between only two species. Examination of the confusion matrix showed that the model correctly classified 32 of 35 amerob recordings and 30 of 34 houspa recordings. Only seven total classification errors were made, indicating that the model learned meaningful differences between the two bird species.

The external audio predictions also highlighted limitations of the multi-class model. All three external recordings were classified as houspa, but the confidence scores varied considerably, ranging from approximately 0.28 to 0.64. While the model was reasonably confident in test2.mp3, it was less certain when classifying test1.mp3 and test3.mp3. Visual inspection of the spectrograms revealed varying acoustic structures and background noise patterns that may differ from those seen during training. These differences likely contributed to the lower confidence scores observed for some recordings.

Several alternative approaches could have been explored for this problem. Transfer learning using larger pre-trained image models such as ResNet or EfficientNet may improve performance by leveraging features learned from large image datasets. Recurrent neural networks (RNNs) or Long Short-Term Memory (LSTM) networks could also be considered because bird calls are inherently sequential audio signals. Additionally, transformer-based audio classification models have recently shown strong performance on many sound recognition tasks.

Despite these alternatives, neural networks remain a logical choice for this application. Spectrograms contain complex visual patterns that can be difficult to represent using manually engineered features. CNNs automatically learn hierarchical representations, beginning with simple frequency patterns and gradually learning more complex acoustic structures. This ability to automatically extract features makes neural networks particularly well suited for bird call classification tasks.

Conclusion

This research investigated the use of convolutional neural networks for bird species classification using spectrogram representations of audio recordings. A multi-class CNN was developed to classify twelve bird species, while a binary CNN was trained to distinguish between amerob and houspa.

The stable multi-class CNN achieved a test accuracy of 42.57%, outperforming the baseline CNN and demonstrating that convolutional neural networks can learn meaningful patterns from bird call spectrograms. The binary CNN achieved substantially stronger performance, reaching 89.86% test accuracy. These results suggest that distinguishing between a small number of species is considerably easier than identifying among many similar bird species.

The external audio predictions further demonstrated both the potential and limitations of the model. Although all three external recordings were classified as houspa, the varying confidence scores suggest that the model was less certain when evaluating recordings that were not part of the training dataset. While the CNN was able to generate species predictions for previously unseen recordings, the confidence values indicate that additional training data and more advanced architectures may improve robustness on unseen audio samples.

Bird sound classification has practical applications in wildlife monitoring, biodiversity conservation, and ecological research. Automated systems capable of identifying bird species from audio recordings could help researchers monitor bird populations more efficiently and at larger scales than manual observation alone. Although additional improvements are needed to increase classification accuracy, the results of this research demonstrate that deep learning provides a promising foundation for automated bird species recognition.

Citation

Vopani. "Xeno-Canto Bird Recordings Extended (A-M)." Kaggle.com, 2020,
www.kaggle.com/datasets/rohanrao/xeno-canto-bird-recordings-extended-a-m.